

RAL-MoE-ROM: HIERARCHICAL MIXTURE-OF-EXPERTS FOR REGIME-ADAPTIVE REDUCED DYNAMICS

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Paper under double-blind review

ABSTRACT

Learning reduced dynamical models over broad parameter ranges is challenging when the underlying system exhibits qualitatively different regimes. A single global representation must simultaneously encode distinct local state geometries, while a single learned dynamical model must reproduce different stability, growth, and saturation mechanisms under recursive rollout. We introduce RAL-MoE-ROM, a two-level hierarchical mixture-of-experts framework for regime-adaptive reduced dynamics. The parameterized solution set is represented by overlapping regime-local reduced charts, each equipped with an autonomous physics–data specialist. Within each specialist, a structured sparse mixture of experts learns corrections to the projected reduced dynamics using nonlinear, affine, and low-rank bilinear expert branches selected by hierarchical sparse routing. A second routing level operates across regime-local specialists. Because the local charts have independent reduced coordinates, adjacent specialists are advanced independently and combined only after reconstruction in the common physical space through a learned Top-2 convex correction. Across three parametrized incompressible-flow benchmarks spanning steady, Hopf-onset, and periodic dynamics, the proposed specialists provide accurate long-horizon autonomous predictions, while the hierarchical fusion consistently improves predictions in evaluated transition regions. Ablations further show complementary contributions from the structured expert parameterization, the projected physical backbone, regime localization, and learned cross-regime routing.

1 INTRODUCTION

Learning compact dynamical models is central to scientific machine learning when high-fidelity simulation is too expensive for repeated evaluation. The problem is especially difficult when a physical parameter changes the qualitative dynamics, not only the response magnitude. Parameterized incompressible flows are representative: different parts of the parameter domain may relax to a steady state, exhibit weak oscillatory growth near instability onset, or sustain nonlinear periodic motion. A many-query reduced model must remain accurate under autonomous rollout across all of these regimes. Unlike a snapshot interpolant, it must continue evolving after the short initial history is exhausted, without injecting reference states into the prediction window.

Broad parameter ranges create two coupled difficulties. First, the state representation is regime dependent: a basis that efficiently describes nearly steady states need not compactly represent developed periodic dynamics. Second, the reduced evolution law must reproduce distinct mechanisms such as decay, growth, phase evolution, and saturation. A global model must solve both problems simultaneously, while recursive prediction continually feeds its own errors back into the dynamics. The difficulty is therefore not only model capacity: global coordinates can spend dimensions reconciling distinct state geometries, and one global evolution law can average mechanisms that should remain regime specific.

Projection-based reduced-order models provide explicit low-dimensional structure through POD and Galerkin projection (Sirovich, 1987; Berkooz et al., 1993; Benner et al., 2015), but truncated interactions can degrade closure and long-horizon behavior (Amsallem & Farhat, 2012). Learned

dynamics offer richer nonlinear maps, yet one-step accuracy need not survive recursive use (Pathak et al., 2018; Vlachas et al., 2018; Duraisamy et al., 2019; Brunton et al., 2020). Hybrid methods therefore combine a projected backbone with learned unresolved terms (Dar et al., 2023; Lee & Carlberg, 2020; Fresca & Manzoni, 2022; Geelen et al., 2023). A complementary approach replaces one global representation with local subspaces, clusters, or manifolds (Kaiser et al., 2014; Li et al., 2021; Diaz et al., 2024). These two remedies address complementary parts of the problem—the first improves the dynamical inductive bias, while the second reduces representation mismatch.

Localization, however, creates an assembly problem. Independently constructed charts have different affine means, bases, scalings, and reduced operators, so their coefficient vectors have no canonical componentwise correspondence. Switching or averaging those vectors can therefore mix incompatible representations. Mixture-of-experts (MoE) models provide a natural language for conditional computation (Jacobs et al., 1991; Jordan & Jacobs, 1994; Shazeer et al., 2017; Fedus et al., 2022), but conventional MoE layers typically route predictors acting in a shared feature space. Regime-dependent reduced dynamics instead require both conditional closure within a chart and selection or combination of complete autonomous models across noncommensurate charts. The outer decision must also respect which local models are supported near a regime boundary; router confidence alone does not make a cross-chart combination dynamically meaningful.

We introduce the *Regime-Adaptive Localized Mixture-of-Experts Reduced-Order Model* (RAL-MoE-ROM), a two-level hierarchy for this setting. Overlapping regime-local charts each support an autonomous physics–data specialist. Within a specialist, hierarchical sparse routing selects structured correction experts with nonlinear, affine, and low-rank bilinear branches while retaining the projected dynamics as an explicit inductive bias. Group Top-1 and channel Top-2 decisions activate a shared expert and a small routed subset for the velocity and pressure channels. Specialists are trained with closed-loop multi-step objectives so that the learned correction is exposed to its own recursive inputs. Across specialists, a parameter-only E2 router produces one trajectory-level regime preference. Physical-history descriptors are used only by the Top-2 convex correction (T2-C) in an admissible overlap. The two selected specialists encode the same physical history in their native charts, evolve independently, and are combined only after reconstruction on the common physical mesh. The fused field is an output and is never fed back into either local rollout.

Our contributions are threefold:

- **Two-level hierarchical routing.** Sparse expert routing models unresolved dynamics inside each local specialist, while a second level selects or combines complete specialists across regimes. This distinguishes conditional closure from conditional selection of an entire dynamical system.
- **Structured closures for autonomous reduced dynamics.** Nonlinear, affine, and low-rank bilinear expert branches augment an explicit projected backbone and are trained under closed-loop rollout. Separate velocity and pressure routing allows the two algebraically different correction channels to adapt independently.
- **Physical-space assembly of noncommensurate local models.** Adjacent specialists retain independent coordinates and rollouts; T2-C couples only their reconstructed fields, avoiding coefficient or operator interpolation. A learned history-conditioned correction refines the parameter-only routing preference while preserving a convex output.

We evaluate RAL-MoE-ROM on three parameterized incompressible-flow systems spanning steady, Hopf-onset, and periodic dynamics. The experiments separate the effects of the structured expert map, projected backbone, localization, and cross-regime assembly. Over the evaluated finite horizons, the local specialists remain accurate and finite, and T2-C improves the corresponding hard or single-specialist prediction in the tested overlaps. The comparisons include vanilla experts, data-only dynamics, a global chart, POD–Galerkin, one-step POD–DeepONet, hard routing, and fixed or probability-based blends. Localization is not uniformly superior to a global model at every parameter; its value emerges from combining local coordinates, regime-specific dynamics, and adaptive assembly.

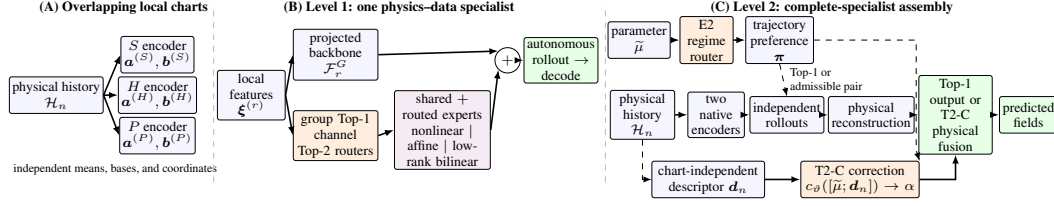


Figure 1: Overview of RAL-MoE-ROM. Sparse routing acts within each regime-local specialist, while parameter-only regime routing selects complete local models. Physical-history descriptors condition only T2-C, which combines independently evolved specialists after reconstruction in the common physical space.

2 RELATED WORK

Projection, learned dynamics, and autonomous rollout. POD–Galerkin methods project governing equations onto a data-adapted basis and retain an interpretable reduced vector field (Sirovich, 1987; Berkooz et al., 1993; Benner et al., 2015). Their long-horizon accuracy can be limited by truncated scales, imperfect closure, and reduced model stabilization (Amsallem & Farhat, 2012). Non-intrusive models instead learn reduced or latent dynamics (Hesthaven & Ubbiali, 2018; Lee & Carlberg, 2020; Champion et al., 2019; Park et al., 2024), and autoregressive forecasters demonstrate the expressivity of learned evolution laws for complex temporal systems (Pathak et al., 2018; Vlachas et al., 2018). Neural operators learn maps between function spaces (Lu et al., 2021; Kovachki et al., 2023), whereas hybrid ROMs retain physical or projection structure while learning unresolved components (Duraissamy et al., 2019; Brunton et al., 2020; Dar et al., 2023; Fresca & Manzoni, 2022; Geelen et al., 2023). RAL-MoE-ROM follows the hybrid view, but replaces a single residual map with a conditionally sparse correction trained under closed-loop rollout. This distinction is important because low one-step error alone does not establish reliable recursive prediction.

Localized representations and conditional dynamics. Local reduced bases, operator localization, clustering, and nonlinear local manifolds reduce the representational burden imposed on a single global chart (Amsallem & Farhat, 2008; Peherstorfer et al., 2014; Kaiser et al., 2014; Li et al., 2021; Diaz et al., 2024). These approaches motivate parameter-local representations, but localization also makes the dynamics chart dependent: independently constructed models generally differ in their means, bases, scaling, and operators. RAL-MoE-ROM treats each chart as a complete autonomous specialist and does not assume that two local coefficient vectors are commensurate. Its cross-regime problem is therefore not merely basis selection; it is conditional assembly of independently evolved dynamical systems.

Mixture-of-experts and the cross-chart gap. Classical hierarchical mixtures learn conditional expert assignments (Jacobs et al., 1991; Jordan & Jacobs, 1994), and sparse MoE architectures scale conditional computation by activating only a subset of experts (Shazeer et al., 2017; Fedus et al., 2022). In most such models, the routed predictors consume and produce a shared representation. RAL-MoE-ROM uses two different routing levels. Inside a local chart, group and channel routers activate structured closure experts that share the same reduced coordinates. Outside the chart, the E2 router selects complete autonomous specialists whose coordinates are unrelated. An admissible adjacent pair is never averaged in coefficient space: both models roll out independently and T2-C fuses only their reconstructed physical fields. This separation of within-chart conditional closure from cross-chart dynamical assembly is the central gap addressed here.

3 RAL-MoE-ROM

We consider parameterized dynamics whose trajectories occupy qualitatively different regions of state space as the physical parameter μ varies. RAL-MoE-ROM separates representation, within-regime closure, and cross-regime assembly instead of forcing every trajectory into one latent coordinate system.

3.1 OVERLAPPING REGIME-LOCAL REDUCED CHARTS

Let $r \in \{S, H, P\}$ index steady, Hopf-onset, and periodic regimes, and let $\mathcal{M}_r$ denote overlapping parameter regions. Each chart has its own affine mean, velocity and pressure bases, scaling, and reduced operators. A physical state is encoded and reconstructed as

$$\mathbf{a}^{(r)} = \Phi_{u,r}^\top \mathbf{W}_u(\mathbf{u} - \bar{\mathbf{u}}_r), \quad \hat{\mathbf{u}}^{(r)} = \bar{\mathbf{u}}_r + \Phi_{u,r} \mathbf{a}^{(r)}, \quad (1)$$

with the analogous map for gauge-corrected pressure coefficients $\mathbf{b}^{(r)}$. Because these coordinates are constructed independently, components of $\mathbf{a}^{(r)}$ and $\mathbf{a}^{(s)}$ are not assumed to correspond when $r \neq s$. This is the reason cross-regime fusion is deferred to physical space.

3.2 PHYSICS-DATA SPECIALISTS WITH STRUCTURED SPARSE EXPERTS

Projection supplies a regime-local dynamical backbone,

$$\dot{\mathbf{a}}^{(r)} = \underbrace{\mathbf{c}_r + \mathbf{A}_r \mathbf{a}^{(r)} + \mathbf{H}_r : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}) + \mathbf{C}_r^p \mathbf{b}^{(r)}}_{\mathcal{F}_r^G(\mathbf{a}^{(r)}, \mathbf{b}^{(r)}; \mu)} + \rho_{\theta_r}^u(\boldsymbol{\xi}^{(r)}). \quad (2)$$

The feature vector $\boldsymbol{\xi}^{(r)}$ contains standardized local coordinates, the projected right-hand side, the parameter, and a short state history. The learned correction is a channel-wise sparse MoE,

$$\mathcal{M}_{\theta_r}^c(\boldsymbol{\xi}) = \omega_0^c E_{0,g^*}^c(\boldsymbol{\xi}) + \sum_{e \in \mathcal{T}^c} \omega_e^c E_{e,g^*}^c(\boldsymbol{\xi}), \quad E(\boldsymbol{\xi}) = N(\boldsymbol{\xi}) + W_{\text{lin}} \boldsymbol{\xi} + \kappa W_o[(U^\top \boldsymbol{\xi}) \odot (V^\top \boldsymbol{\xi})]. \quad (3)$$

The first routing decision selects a sparse expert group and the second selects experts within that group. The nonlinear, affine, and low-rank bilinear branches provide complementary correction structures without removing the explicit projected term. Pressure is reconstructed through a physics-data algebraic map rather than evolved as an unconstrained recurrent state; details are in Appendix C.

3.3 CLOSED-LOOP AUTONOMOUS TRAINING

Each specialist is trained on rollout windows. Starting from encoded physical history, the numerical integrator repeatedly evaluates Eq. (2) on its own predictions. The loss aggregates velocity- and pressure-coordinate errors across the window, so the optimization sees the distribution shift produced by recursive use. All comparisons use held-out trajectories and the reported rollout horizons; chart-wise trajectory partitions are listed in Table 5.

3.4 REGIME ROUTING AND PHYSICAL-SPACE T2-C FUSION

The outer E2 router depends only on the standardized physical parameter $\tilde{\mu}$ and produces a trajectory-level regime preference $\pi(\mu) = \text{softmax}(\mathcal{R}_{E2}(\tilde{\mu}))$. Its probabilities are evaluated once and remain fixed during the rollout. Chart-independent descriptors extracted from the initial physical history are used only by the T2-C correction gate in an admissible overlap; they are not E2 inputs. Hard Top-1 prediction advances the largest-probability admissible specialist. When an adjacent pair is admissible ($S-H$ or $H-P$), both specialists encode the same physical history in their native coordinates and then evolve independently. For reconstructed fields $\hat{\mathbf{q}}^{(r)}$ and $\hat{\mathbf{q}}^{(s)}$, T2-C predicts a window-conditioned convex weight and forms

$$\hat{\mathbf{q}}_{\text{T2C}}(t) = \alpha \hat{\mathbf{q}}^{(r)}(t) + (1 - \alpha) \hat{\mathbf{q}}^{(s)}(t), \quad \alpha = \sigma(\text{logit}(\bar{\pi}_r) + \Delta_\psi). \quad (4)$$

Here $\bar{\pi}_r$ is the pair-normalized E2 probability and $\Delta_\psi = c_\psi([\tilde{\mu}; \mathbf{d}_n])$ is a learned correction based on the standardized parameter and chart-independent physical-history descriptor $\mathbf{d}_n$. Fusion occurs after reconstruction and is not fed back into either local rollout. Admissibility is selected from validation support and should not be interpreted as a formal stability certificate.

4 EXPERIMENTS

We evaluate the two levels of RAL-MoE-ROM through complementary benchmark roles. All three two-dimensional incompressible-flow datasets span steady, Hopf-onset, and periodic dynamics as

Table 1: Experimental scope and primary benchmark roles. Local ranks list $(r_u, r_p)_S$, $(r_u, r_p)_H$, and $(r_u, r_p)_P$, in that order; horizons give $K_S/K_H/K_P$. Here K counts evaluation rollout steps at the corresponding sampling interval. Additional overlap and periodic-core settings are identified in the corresponding comparisons.

Benchmark	Re range	Local ranks (S, H, P)	Main horizons	Primary role
Circular cylinder	20–200	(32, 32), (32, 32), (32, 32)	56/56/48	Local specialist ablation
Centered square	50–150	(5, 4), (11, 11), (28, 26)	16/48/24	Routing and fusion
Fluidic pinball	10–60	(4, 3), (5, 5), (17, 16)	56/56/32	Autonomous baseline

Table 2: Autonomous local-specialist prediction on the fluidic-pinball benchmark. Errors are percentages; lower is better. Periodic core is a second evaluation setting within the periodic regime.

Evaluation setting	Model	E_u	E_p	E_{joint}
Steady ($K = 56$)	POD–Galerkin	N/E [†]	N/E [†]	N/E [†]
	RAL–MoE–ROM	0.2247	0.7593	0.9840
Hopf ($K = 56$)	POD–Galerkin	0.4378 [‡]	48.50 [‡]	48.94 [‡]
	RAL–MoE–ROM	0.0995	1.986	2.085
Periodic ($K = 32$)	POD–Galerkin	1.641	41.56	43.20
	RAL–MoE–ROM	0.2897	0.5518	0.8415
Periodic core ($K = 56$)	POD–Galerkin	2.555	39.36	41.91
	RAL–MoE–ROM	0.6222	1.050	1.672

[†] All 448 evaluation windows violate the predefined rollout-divergence criterion.

[‡] Errors are computed on finite evaluation windows, including threshold-violating windows; 141 of 320 Hopf windows violate the divergence criterion. All RAL–MoE–ROM settings and both periodic POD–Galerkin settings have zero divergent windows.

the Reynolds number varies. The circular-cylinder benchmark provides the local-specialist ablation, the centered-square benchmark provides matched S – H and H – P routing/fusion comparisons, and the fluidic-pinball benchmark provides the autonomous comparison with a classical POD–Galerkin baseline. Table 1 summarizes this scope. All training, validation, and test partitions are defined by complete parameter trajectories; chart-wise ranges and split counts are reported in Appendix E.

Errors E_u and E_p are relative physical-field errors in percent, aggregated according to each benchmark’s evaluation protocol; $E_{\text{joint}} = E_u + E_p$. The sum of displayed component errors can differ slightly from the reported joint error because values are rounded independently. “Divergent” denotes an evaluation window that violates the predefined rollout-divergence criterion, including non-finite states or threshold violations. N/E denotes not evaluable under the prescribed rollout criterion; table-specific notes give the reason.

4.1 DO LOCAL SPECIALISTS SUPPORT LONG AUTONOMOUS ROLLOUTS?

The learned fluidic-pinball specialists complete all four evaluation settings without divergent windows, whereas POD–Galerkin fails in every steady window and in part of the Hopf evaluation (Table 2). The Hopf baseline has a joint error of 48.94% on finite windows, compared with 2.085% for RAL–MoE–ROM; at periodic $K = 32$, the joint error decreases from 43.20% to 0.8415%. The separate velocity and pressure columns show the particularly large pressure discrepancy of the projected baseline. These results support autonomous prediction over the evaluated horizons. The circular-cylinder ablation and centered-square local-specialist results provide complementary evidence in Table 4 and Appendix G, respectively.

Table 3: Cross-regime routing and fusion on the centered-square benchmark at $K = 24$. Entries are E_{joint} (%); lower is better. Best single is the lowest-error fixed specialist over each overlap’s evaluation set (Hopf for $S-H$, Periodic for $H-P$), not a window-wise selection. The convex oracle is an a posteriori reference unavailable during prediction; bold marks the best deployable assembly.

Overlap	Best single	E2 Top-1	Equal	E2 prob.	T2-C	Oracle
$S-H$	1.7298	1.7298	3.0321	2.1847	1.1958	1.1842
$H-P$	1.2079	1.8473	1.6146	1.6618	0.8465	0.8384

Table 4: Local architecture ablation on the circular-cylinder benchmark. Each regime cell reports the min-max E_u range (upper line) and E_p range (lower line) over held-out Reynolds-number trajectories, in percent. These are field-error ranges, not means or uncertainty intervals. N/E denotes not evaluable under the prescribed rollout criterion: the Vanilla-FNN-MoE Hopf model does not meet the validation-stage stability requirement and is not evaluated on held-out trajectories.

Model	Steady ($K = 56$)	Hopf ($K = 56$)	Periodic ($K = 48$)
RAL-MoE-ROM	0.0330–0.1723	0.0032–0.0288	0.3809–1.3473
	0.2500–1.3249	0.0355–0.4321	1.6825–5.9601
Vanilla-FNN-MoE	0.0384–0.2995	N/E	1.0393–3.0836
	2.2163–18.5603		3.9659–13.3257
DataOnly-MoE	0.1637–0.5122	0.1260–0.3880	0.7854–1.8677
	4751.1025–27275.4645	1280.2811–1524.5949	2.2511–10.8189
Global MoE	0.0845–0.3404	0.0110–0.0268	0.5873–1.1726
	1.5226–7.4363	0.0560–0.2882	2.1896–6.3566

4.2 DOES LEARNED PHYSICAL-SPACE FUSION IMPROVE OVERLAP PREDICTION?

Having evaluated autonomous local dynamics, we next isolate the second level of the hierarchy. The centered-square benchmark provides both $S-H$ and $H-P$ overlaps under a common $K = 24$ protocol and is therefore the primary routing/fusion comparison. Table 3 reports joint field error; separate velocity and pressure errors are given in Table 7 in Appendix G.

T2-C improves both hard routing and the best fixed individual specialist in the two tested overlaps. In $H-P$, joint error decreases from 1.2079% for the best single specialist and 1.8473% for E2 Top-1 to 0.8465%. Its proximity to the convex oracle suggests that the learned correction captures most of the available complementarity within the tested convex fusion class. Figure 2 shows representative physical fields and velocity-error maps; parameter-wise results and the fluidic-pinball overlap study appear in Appendices G and H.

4.3 WHAT DO LOCAL ARCHITECTURAL COMPONENTS CONTRIBUTE?

The circular-cylinder comparison isolates three local modeling choices (Table 4). Vanilla-FNN-MoE replaces the structured expert map with conventional feed-forward experts, DataOnly-MoE removes the projected Galerkin and Pressure–Poisson contributions, and Global MoE replaces the local charts and specialists with one global model. The comparison uses physical-field errors so that models with different reduced coordinates can be evaluated on a common quantity.

Structured specialists improve the steady and periodic error ranges over Vanilla-FNN-MoE, which does not meet the validation-stage stability requirement for the Hopf evaluation. For example, the steady pressure range is 0.2500–1.3249% for RAL-MoE-ROM versus 2.2163–18.5603% for Vanilla-FNN-MoE. Removing the physical backbone particularly degrades steady and Hopf pressure prediction. The global model remains competitive in Hopf and periodic physical-field errors, so localization is not uniformly superior; its role is to provide regime-specific coordinates and dynamics that can participate in cross-regime assembly. The local comparisons address expert structure, physics, and localization, while Table 3 separately evaluates the outer assembly.

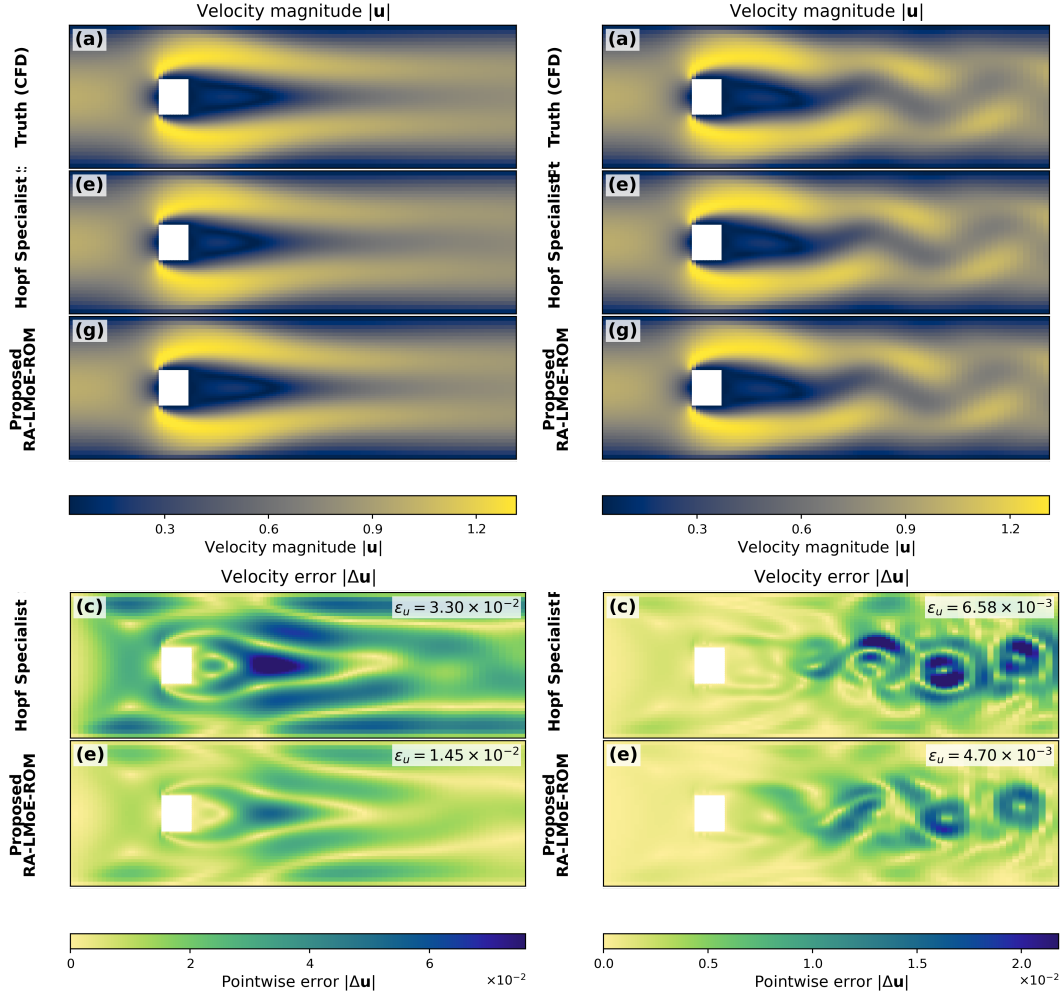


Figure 2: Representative cross-regime predictions on the centered-square benchmark. The left and right panels correspond to the $S-H$ and $H-P$ overlaps, respectively. T2-C combines independently evolved specialists after physical reconstruction and reduces the velocity-field discrepancy relative to the corresponding single-specialist prediction. Complete velocity and pressure fields are reported in Appendix G.

5 LIMITATIONS

The regime cover and admissible adjacency graph are selected using validation support; the method does not discover arbitrary bifurcation structure or certify unseen transitions. T2-C is evaluated only between adjacent specialists, and convex physical-space fusion may be insufficient when more than two regimes interact or when conserved quantities require a constrained assembly rule. The evidence concerns two-dimensional incompressible flows and empirical finite-horizon behavior; it does not establish provable stability, conservation, or accuracy outside the sampled parameter ranges. A separate hierarchy is trained for each benchmark, so cross-geometry transfer of one hierarchy is not evaluated. Multiple local bases, operators, and specialists also increase training and model-management cost relative to one global ROM.

The current evaluation is based on fixed validation-selected training runs; repeated-seed uncertainty and parameter-matched capacity comparisons are left for future evaluation.

6 CONCLUSION

RAL-MoE-ROM casts broad-parameter reduced dynamics as a two-level mixture-of-experts problem. Structured sparse experts learn unresolved dynamics inside each physics–data specialist, while an outer router selects or combines complete local models. Keeping the local coordinates independent and coupling only reconstructed physical fields avoids interpolation between incompatible modal states. Across the three evaluated flow systems, the specialists achieve low error over the evaluated autonomous rollout horizons, and T2-C improves prediction in the tested transition regions. The results motivate extensions that learn the regime cover, impose physical constraints on field-space fusion, and evaluate the hierarchy on higher-dimensional and non-fluid dynamical systems.

REPRODUCIBILITY STATEMENT

The main text reports the benchmark roles, evaluation horizons, baselines, and metrics. Appendices A–E document the local-coordinate construction, projected operators, specialist architecture, routing and fusion procedure, and implementation details. Appendices F–H provide benchmark-specific results.

AI USE STATEMENT

TODO: authors to complete final disclosure.

REFERENCES

- David Amsallem and Charbel Farhat. Interpolation method for adapting reduced-order models and application to aeroelasticity. *AIAA Journal*, 46(7):1803–1813, 2008. doi: 10.2514/1.35374.
- David Amsallem and Charbel Farhat. Stabilization of projection-based reduced-order models. *International Journal for Numerical Methods in Engineering*, 91(4):358–377, 2012. doi: 10.1002/nme.4274.
- Peter Benner, Serkan Gugercin, and Karen Willcox. A survey of projection-based model reduction methods for parametric dynamical systems. *SIAM Review*, 57(4):483–531, 2015. doi: 10.1137/130932715.
- Gal Berkooz, Philip Holmes, and John L. Lumley. The proper orthogonal decomposition in the analysis of turbulent flows. *Annual Review of Fluid Mechanics*, 25:539–575, 1993. doi: 10.1146/annurev.fl.25.010193.002543.
- Steven L. Brunton, Bernd R. Noack, and Petros Koumoutsakos. Machine learning for fluid mechanics. *Annual Review of Fluid Mechanics*, 52:477–508, 2020. doi: 10.1146/annurev-fluid-010719-060214.
- Kathleen Champion, Bethany Lusch, J. Nathan Kutz, and Steven L. Brunton. Data-driven discovery of coordinates and governing equations. *Proceedings of the National Academy of Sciences*, 116(45):22445–22451, 2019. doi: 10.1073/pnas.1906995116.
- Zulkeefal Dar, Joan Baiges, and Ramon Codina. Artificial neural network based correction for reduced order models in computational fluid mechanics. *Computer Methods in Applied Mechanics and Engineering*, 415:116232, 2023. doi: 10.1016/j.cma.2023.116232.
- Alejandro N. Diaz, Youngsoo Choi, and Matthias Heinkenschloss. A fast and accurate domain decomposition nonlinear manifold reduced order model. *Computer Methods in Applied Mechanics and Engineering*, 425:116943, 2024. doi: 10.1016/j.cma.2024.116943.
- Karthik Duraisamy, Gianluca Iaccarino, and Heng Xiao. Turbulence modeling in the age of data. *Annual Review of Fluid Mechanics*, 51:357–377, 2019. doi: 10.1146/annurev-fluid-010518-040547.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022. URL <https://jmlr.org/papers/v23/21-0998.html>.

- Stefania Fresca and Andrea Manzoni. POD-DL-ROM: Enhancing deep learning-based reduced order models for nonlinear parametrized PDEs by proper orthogonal decomposition. *Computer Methods in Applied Mechanics and Engineering*, 388:114181, 2022. doi: 10.1016/j.cma.2021.114181.
- Rudy Geelen, Stephen J. Wright, and Karen E. Willcox. Operator inference for non-intrusive model reduction with quadratic manifolds. *Computer Methods in Applied Mechanics and Engineering*, 403:115717, 2023. doi: 10.1016/j.cma.2022.115717.
- Jan S. Hesthaven and Stefano Ubbiali. Non-intrusive reduced order modeling of nonlinear problems using neural networks. *Journal of Computational Physics*, 363:55–78, 2018. doi: 10.1016/j.jcp.2018.02.037.
- Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. Adaptive mixtures of local experts. *Neural Computation*, 3(1):79–87, 1991. doi: 10.1162/neco.1991.3.1.79.
- Michael I. Jordan and Robert A. Jacobs. Hierarchical mixtures of experts and the EM algorithm. *Neural Computation*, 6(2):181–214, 1994. doi: 10.1162/neco.1994.6.2.181.
- Eurika Kaiser, Bernd R. Noack, Laurent Cordier, Andreas Spohn, Marc Segond, Markus Abel, Guillaume Daviller, Jan Östh, Siniša Krajnović, and Robert K. Niven. Cluster-based reduced-order modelling of a mixing layer. *Journal of Fluid Mechanics*, 754:365–414, 2014. doi: 10.1017/jfm.2014.355.
- Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023. URL <https://jmlr.org/papers/v24/21-1524.html>.
- Kookjin Lee and Kevin T. Carlberg. Model reduction of dynamical systems on nonlinear manifolds using deep convolutional autoencoders. *Journal of Computational Physics*, 404:108973, 2020. doi: 10.1016/j.jcp.2019.108973.
- Hao Li, Daniel Fernex, Richard Semaan, Jun Tan, Marek Morzyński, and Bernd R. Noack. Cluster-based network model. *Science Advances*, 7(25):eabf5006, 2021. doi: 10.1126/sciadv.abf5006.
- Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021. doi: 10.1038/s42256-021-00302-5.
- Jun Sur Richard Park, Siu Wun Cheung, Youngsoo Choi, and Yeonjong Shin. tLaSDI: Thermodynamics-informed latent space dynamics identification. *Computer Methods in Applied Mechanics and Engineering*, 429:117144, 2024. doi: 10.1016/j.cma.2024.117144.
- Jaideep Pathak, Brian Hunt, Michelle Girvan, Zhixin Lu, and Edward Ott. Model-free prediction of large spatiotemporally chaotic systems from data: A reservoir computing approach. *Physical Review Letters*, 120:024102, 2018. doi: 10.1103/PhysRevLett.120.024102.
- Benjamin Peherstorfer, Daniel Butnaru, Karen Willcox, and Hans-Joachim Bungartz. Localized discrete empirical interpolation method. *SIAM Journal on Scientific Computing*, 36(1):A168–A192, 2014. doi: 10.1137/130924408.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*, 2017. URL <https://openreview.net/forum?id=BlckMDqlg>.
- Lawrence Sirovich. Turbulence and the dynamics of coherent structures. part i: Coherent structures. *Quarterly of Applied Mathematics*, 45(3):561–571, 1987. doi: 10.1090/qam/910462.
- Pantelis R. Vlachas, Wonmin Byeon, Zhong Y. Wan, Themistoklis P. Sapsis, and Petros Koumoutsakos. Data-driven forecasting of high-dimensional chaotic systems with long short-term memory networks. *Proceedings of the Royal Society A*, 474(2213):20170844, 2018. doi: 10.1098/rspa.2017.0844.

A FULL PROBLEM SETUP AND WEIGHTED POD CONSTRUCTION

A.1 PARAMETERIZED FULL-ORDER MODEL, GAUGE, AND REGIME COVER

After spatial discretization, the velocity $\mathbf{u}(t; \mu) \in \mathbb{R}^{N_u}$ and pressure $p(t; \mu) \in \mathbb{R}^{N_p}$ satisfy

$$\dot{\mathbf{u}} = \mathbf{f}_h(\mu) + \mathcal{L}_h(\mu)\mathbf{u} - \mathcal{N}_h(\mathbf{u}, \mathbf{u}; \mu) - \nabla_h(\mu)p, \quad \text{div}_h(\mu)\mathbf{u} = 0. \quad (5)$$

Pressure is defined only up to an additive constant. With positive finite-volume weights $\mathbf{w}$, every pressure snapshot is mapped to the weighted zero-mean gauge

$$\Gamma_p(q) = q - \frac{\mathbf{w}^\top q}{\mathbf{w}^\top \mathbf{1}} \mathbf{1}, \quad p^\circ = \Gamma_p(p), \quad \mathbf{w}^\top p^\circ = 0. \quad (6)$$

The parameter domain is covered by overlapping regions

$$\mathcal{M} \subseteq \bigcup_{r \in \{S, H, P\}} \mathcal{M}_r, \quad \mathcal{O}_{SH} = \mathcal{M}_S \cap \mathcal{M}_H, \quad \mathcal{O}_{HP} = \mathcal{M}_H \cap \mathcal{M}_P. \quad (7)$$

No direct S – P overlap is used.

A.2 AREA-WEIGHTED LOCAL POD

The finite-volume quadrature induces

$$\langle \mathbf{v}, \mathbf{z} \rangle_{M_u} = \mathbf{v}^\top M_u \mathbf{z}, \quad \langle q, s \rangle_{M_p} = q^\top M_p s, \quad M_p = \text{diag}(\mathbf{w}), \quad M_u = \text{diag}(M_p, M_p). \quad (8)$$

For chart r , let $X_u^{(r)}$ and $X_p^{(r)}$ contain velocity and gauge-corrected pressure snapshots after subtracting their chart-specific means $\bar{\mathbf{u}}^{(r)}$ and $\bar{p}^{(r), \circ}$. The local bases solve

$$\Phi_u^{(r)} \in \arg \min_{\Psi^\top M_u \Psi = I_{d_u}} \left\| X_u^{(r)} - \Psi \Psi^\top M_u X_u^{(r)} \right\|_{M_u, F}^2, \quad (9)$$

$$\Phi_p^{(r)} \in \arg \min_{\Psi^\top M_p \Psi = I_{d_p}} \left\| X_p^{(r)} - \Psi \Psi^\top M_p X_p^{(r)} \right\|_{M_p, F}^2, \quad (10)$$

where $\|Y\|_{M, F}^2 = \text{tr}(Y^\top M Y)$. Equivalently, for the weighted singular value decompositions

$$M_u^{1/2} X_u^{(r)} = U_u^{(r)} \Sigma_u^{(r)} V_u^{(r)\top}, \quad M_p^{1/2} X_p^{(r)} = U_p^{(r)} \Sigma_p^{(r)} V_p^{(r)\top}, \quad (11)$$

the bases are

$$\Phi_u^{(r)} = M_u^{-1/2} U_u^{(r)}(:, 1:d_u), \quad \Phi_p^{(r)} = M_p^{-1/2} U_p^{(r)}(:, 1:d_p), \quad (12)$$

and satisfy $\Phi_u^{(r)\top} M_u \Phi_u^{(r)} = I$ and $\Phi_p^{(r)\top} M_p \Phi_p^{(r)} = I$. Gauge consistency also gives $\mathbf{w}^\top \Phi_p^{(r)} = \mathbf{0}^\top$.

Encoding and decoding are chart specific:

$$\mathbf{a}^{(r)} = \Phi_u^{(r)\top} M_u (\mathbf{u} - \bar{\mathbf{u}}^{(r)}), \quad \mathcal{D}_r^u(\mathbf{a}^{(r)}) = \bar{\mathbf{u}}^{(r)} + \Phi_u^{(r)} \mathbf{a}^{(r)}, \quad (13)$$

$$\mathbf{b}^{(r)} = \Phi_p^{(r)\top} M_p (\Gamma_p(p) - \bar{p}^{(r), \circ}), \quad \mathcal{D}_r^p(\mathbf{b}^{(r)}) = \bar{p}^{(r), \circ} + \Phi_p^{(r)} \mathbf{b}^{(r)}. \quad (14)$$

Learned components use local standardizations

$$\tilde{\mathbf{a}}^{(r)} = (\mathbf{a}^{(r)} - \boldsymbol{\mu}_a^{(r)}) \oslash \boldsymbol{\sigma}_a^{(r)}, \quad \tilde{\mathbf{b}}^{(r)} = (\mathbf{b}^{(r)} - \boldsymbol{\mu}_b^{(r)}) \oslash \boldsymbol{\sigma}_b^{(r)}. \quad (15)$$

B PROJECTED GALERKIN AND PRESSURE OPERATORS

B.1 CHART-LOCAL MOMENTUM DYNAMICS

Substituting the affine reconstructions into Eq. (5) and testing against the local velocity basis gives

$$\dot{\mathbf{a}}^{(r)} = \mathbf{c}_r(\mu) + A_r(\mu) \mathbf{a}^{(r)} + H_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}) + C_r^p(\mu) \mathbf{b}^{(r)}. \quad (16)$$

For velocity modes $\phi_i^{u,(r)}$ and pressure modes $\phi_\ell^{p,(r)}$,

$$[c_r]_i = \left\langle \phi_i^{u,(r)}, \mathbf{f}_h + \mathcal{L}_h \bar{\mathbf{u}}^{(r)} - \mathcal{N}_h(\bar{\mathbf{u}}^{(r)}, \bar{\mathbf{u}}^{(r)}; \mu) - \nabla_h \bar{p}^{(r),\circ} \right\rangle_{M_u}, \quad (17)$$

$$[A_r]_{ij} = \left\langle \phi_i^{u,(r)}, \mathcal{L}_h \phi_j^{u,(r)} - \mathcal{N}_h(\bar{\mathbf{u}}^{(r)}, \phi_j^{u,(r)}; \mu) - \mathcal{N}_h(\phi_j^{u,(r)}, \bar{\mathbf{u}}^{(r)}; \mu) \right\rangle_{M_u}, \quad (18)$$

$$[H_r]_{ijk} = - \left\langle \phi_i^{u,(r)}, \mathcal{N}_h(\phi_j^{u,(r)}, \phi_k^{u,(r)}; \mu) \right\rangle_{M_u}, \quad (19)$$

$$[C_r^p]_{i\ell} = - \left\langle \phi_i^{u,(r)}, \nabla_h(\mu) \phi_\ell^{p,(r)} \right\rangle_{M_u}. \quad (20)$$

These terms define the backbone $\mathcal{F}_r^G(\mathbf{a}, \mathbf{b}; \mu)$ used in Eq. (2).

B.2 GAUGE-CONSISTENT PRESSURE-POISSON MAP

At the continuous level the gauge-fixed pressure satisfies

$$-\Delta p^\circ = \nabla \cdot [(\mathbf{u} \cdot \nabla) \mathbf{u} - \mathbf{f}], \quad \int_{\Omega} p^\circ \, d\Omega = 0, \quad (21)$$

with boundary conditions inherited from the momentum equation. Projection of the discrete relation gives

$$K_r^p(\mu) \mathbf{b}^{PP,(r)} = \mathbf{d}_r(\mu) + B_r(\mu) \mathbf{a}^{(r)} + Q_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}), \quad (22)$$

and hence

$$\mathcal{Q}_r^{PP}(\mathbf{a}; \mu) = [K_r^p(\mu)]^{-1} [\mathbf{d}_r(\mu) + B_r(\mu) \mathbf{a} + Q_r(\mu) : (\mathbf{a} \otimes \mathbf{a})]. \quad (23)$$

The gauge removes the constant pressure nullspace. Each complete chart therefore contains its own parameter region, affine means, bases, scalars, $\mathcal{F}_r^G$, and $\mathcal{Q}_r^{PP}$. Equal reduced dimensions do not create a canonical componentwise identification between two charts; only decoded physical fields share a common representation.

C SPECIALIST ARCHITECTURE AND CLOSED-LOOP OBJECTIVE

Each regime-local chart is equipped with an independently parameterized semi-explicit specialist. The velocity coefficients evolve under the sum of the chart-local Galerkin vector field and a learned closure,

$$\dot{\mathbf{a}}^{(r)} = \mathcal{F}_r^G(\mathbf{a}^{(r)}, \mathbf{b}^{(r)}; \mu) + \rho_{\theta_r}^u(\boldsymbol{\xi}^{(r)}). \quad (24)$$

Pressure is algebraic rather than an independently evolved recurrent state. After the velocity macro-step, the local pressure-Poisson operator provides the physics-based candidate

$$\mathbf{b}_{n+1}^{PP,(r)} = \mathcal{Q}_r^{PP}(\mathbf{a}_{n+1}^{(r)}; \mu), \quad (25)$$

while a data-driven candidate and a componentwise gate are evaluated from the specialist feature vector,

$$\hat{\mathbf{b}}_{n+1}^{D,(r)} = \rho_{\theta_r}^p(\boldsymbol{\xi}_n^{(r)}), \quad \boldsymbol{\alpha}_n^{(r)} = \text{sigmoid}(\boldsymbol{\eta}_{\theta_r}^p(\boldsymbol{\xi}_n^{(r)})). \quad (26)$$

The next pressure state is the componentwise convex combination

$$\mathbf{b}_{n+1}^{(r)} = \boldsymbol{\alpha}_n^{(r)} \odot \mathbf{b}_{n+1}^{PP,(r)} + (1 - \boldsymbol{\alpha}_n^{(r)}) \odot \hat{\mathbf{b}}_{n+1}^{D,(r)}, \quad \boldsymbol{\alpha}_n^{(r)} \in (0, 1)^{d_p}. \quad (27)$$

C.1 FEATURES AND STRUCTURED SPARSE EXPERTS

Let

$$\mathbf{g}_n^{(r)} = \mathcal{F}_r^G(\mathbf{a}_n^{(r)}, \mathbf{b}_n^{(r)}; \mu) \quad (28)$$

be the current projected right-hand side, and let tildes denote standardization with statistics from the corresponding regime-local training population. The current and two preceding feature blocks are

$$\chi_n^{(r)} = \text{col}\left(\tilde{\mathbf{a}}_n^{(r)}, \tilde{\mathbf{b}}_n^{(r)}, \tilde{\mathbf{g}}_n^{(r)}, \tilde{\mu}\right), \quad (29)$$

$$\mathbf{h}_{n,\ell}^{(r)} = \text{col}\left(\tilde{\mathbf{a}}_{n-\ell}^{(r)}, \tilde{\mathbf{b}}_{n-\ell}^{(r)}, \tilde{\mathbf{g}}_{n-\ell}^{(r)}, \tilde{\mu}\right), \quad \ell \in \{1, 2\}. \quad (30)$$

Thus the complete input is the three-state history

$$\boldsymbol{\xi}_n^{(r)} = \text{col}\left(\chi_n^{(r)}, \mathbf{h}_{n,1}^{(r)}, \mathbf{h}_{n,2}^{(r)}\right). \quad (31)$$

The sampling interval is supplied to the numerical integrator; it is not an additional trainable network feature.

Within each specialist, the group router first makes a hard Top-1 decision,

$$g^* = \arg \max_{g \in \mathcal{G}} \pi_g^{\text{grp}}(\boldsymbol{\xi}). \quad (32)$$

Velocity and pressure then use separate channel routers. For $c \in \{u, p\}$, let $\mathcal{T}_2^c(\boldsymbol{\xi}; g^*)$ contain the two routed experts with the largest channel-specific scores. Together with the shared expert $e = 0$, the active output is

$$\mathcal{M}_{\theta_r}^c(\boldsymbol{\xi}) = \omega_0^c E_{0,g^*}^c(\boldsymbol{\xi}) + \sum_{e \in \mathcal{T}_2^c(\boldsymbol{\xi}; g^*)} \omega_e^c E_{e,g^*}^c(\boldsymbol{\xi}), \quad \omega_e^c \geq 0, \quad \omega_0^c + \sum_{e \in \mathcal{T}_2^c} \omega_e^c = 1. \quad (33)$$

Consequently, one group, one shared expert, and two routed experts are active for each output channel. Each expert combines nonlinear, linear/affine, and low-rank bilinear branches,

$$E_{e,g}^c(\boldsymbol{\xi}) = N_{e,g}^c(\boldsymbol{\xi}) + W_{e,g}^{c,\text{lin}} \boldsymbol{\xi} + \kappa W_{e,g}^{c,o} \left[\left(U_{e,g}^{c,\top} \boldsymbol{\xi} \right) \odot \left(V_{e,g}^{c,\top} \boldsymbol{\xi} \right) \right], \quad q_b = 4, \quad \kappa = 0.05, \quad (34)$$

where q_b is the rank of the bilinear branch and κ scales its contribution to the structured expert map.

C.2 FROZEN-CONTEXT ADVANCEMENT AND CLOSED-LOOP TRAINING

During a velocity macro-step, the current pressure, the recent history, and the physical parameter form a context $\mathcal{C}_n^{(r)}$ that is held fixed across all Runge–Kutta stages. Define

$$F_r(\mathbf{a}; \mathcal{C}_n^{(r)}) = \mathcal{F}_r^G(\mathbf{a}, \mathbf{b}_n^{(r)}; \mu) + \boldsymbol{\rho}_{\theta_r}^u(\boldsymbol{\xi}^{(r)}(\mathbf{a}; \mathcal{C}_n^{(r)})). \quad (35)$$

All three specialists use the classical fourth-order Runge–Kutta update

$$\begin{aligned} \mathbf{k}_1 &= F_r(\mathbf{a}_n^{(r)}; \mathcal{C}_n^{(r)}), \\ \mathbf{k}_2 &= F_r\left(\mathbf{a}_n^{(r)} + \frac{\Delta t_r(\mu)}{2} \mathbf{k}_1; \mathcal{C}_n^{(r)}\right), \\ \mathbf{k}_3 &= F_r\left(\mathbf{a}_n^{(r)} + \frac{\Delta t_r(\mu)}{2} \mathbf{k}_2; \mathcal{C}_n^{(r)}\right), \\ \mathbf{k}_4 &= F_r\left(\mathbf{a}_n^{(r)} + \Delta t_r(\mu) \mathbf{k}_3; \mathcal{C}_n^{(r)}\right), \\ \mathbf{a}_{n+1}^{(r)} &= \mathbf{a}_n^{(r)} + \frac{\Delta t_r(\mu)}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \end{aligned} \quad (36)$$

The updated velocity is supplied to Eq. (27); the history is then shifted and the autonomous rollout continues.

For a closed-loop rollout of length K , the model recursively consumes its own predicted states and minimizes the regime-locally normalized coefficient error

$$\mathcal{L}_r = \frac{1}{K} \sum_{k=1}^K \left[\frac{1}{d_u} \left\| \left(\hat{\mathbf{a}}_{n+k}^{(r)} - \mathbf{a}_{n+k}^{(r)} \right) \odot \boldsymbol{\sigma}_a^{(r)} \right\|_2^2 + \frac{1}{d_p} \left\| \left(\hat{\mathbf{b}}_{n+k}^{(r)} - \mathbf{b}_{n+k}^{(r)} \right) \odot \boldsymbol{\sigma}_b^{(r)} \right\|_2^2 \right]. \quad (37)$$

Training minimizes Eq. (37) over progressively longer rollout horizons $K \in \{4, 8, 12, 16\}$. The three specialists are trained and selected independently; subsequent outer-router or T2-C training does not update their local bases, scalars, reduced operators, internal routers, experts, integrators, or pressure maps.

D OUTER ROUTING AND T2-C DETAILS

Parameter-only routing and admissibility. The E2 router receives only the standardized scalar parameter,

$$\tilde{\mu} = \frac{\mu - \mu_R}{\sigma_R}, \quad \boldsymbol{\pi}(\mu) = \text{softmax}(\mathcal{R}_{E2}(\tilde{\mu})) = [\pi_S, \pi_H, \pi_P]^\top. \quad (38)$$

The trajectory-level probabilities are evaluated once and remain unchanged during rollout. Adjacency is fixed by

$$A_{\text{adj}} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad \mathcal{C}_{\text{top}}(k) = \{j : A_{\text{adj},kj} = 1\}, \quad (39)$$

with ordering (S, H, P) , so direct S - P fusion is excluded. If $m_j(\mathcal{H}, \mu, K) \in \{0, 1\}$ denotes validation-supported dynamical admissibility, the masked preference is

$$\tilde{\pi}_j = \frac{\mathbf{1}_{\{j \in \mathcal{C}_{\text{top}}\}} m_j \pi_j}{\sum_{k \in \{S, H, P\}} \mathbf{1}_{\{k \in \mathcal{C}_{\text{top}}\}} m_k \pi_k + \varepsilon}. \quad (40)$$

One admissible candidate gives a Top-1 prediction; an admissible adjacent pair is passed to T2-C. No ROM prediction is asserted when every mask is zero. For a pair $(1, 2)$, define

$$\bar{\pi}_1 = \frac{\pi_1}{\pi_1 + \pi_2}, \quad \bar{\pi}_2 = 1 - \bar{\pi}_1, \quad \ell^{E2} = \log \frac{\bar{\pi}_1 + \varepsilon}{\bar{\pi}_2 + \varepsilon}. \quad (41)$$

Chart-independent physical-history descriptor. T2-C, unlike E2, uses the current physical state and two preceding states. Let $|\Omega| = \mathbf{1}^\top \mathbf{w}$, $\Delta t_n = t_n - t_{n-1}$, and let U_{ref} , P_{ref} , and T_{ref} be fixed reference scales. With gauge-corrected pressure p_n° , define

$$\hat{E}_n^u = \frac{\|\mathbf{u}_n\|_{M_u}^2}{|\Omega| U_{\text{ref}}^2}, \quad \hat{E}_n^p = \frac{\|p_n^\circ\|_{M_p}^2}{|\Omega| P_{\text{ref}}^2}, \quad (42)$$

$$\hat{D}_n^u = \frac{T_{\text{ref}} \|\mathbf{u}_n - \mathbf{u}_{n-1}\|_{M_u}}{\Delta t_n \sqrt{|\Omega|} U_{\text{ref}}}, \quad \hat{D}_n^p = \frac{T_{\text{ref}} \|p_n^\circ - p_{n-1}^\circ\|_{M_p}}{\Delta t_n \sqrt{|\Omega|} P_{\text{ref}}}, \quad (43)$$

$$\hat{g}_n = \frac{T_{\text{ref}}}{\Delta t_n} \log \frac{\hat{E}_n^u + \varepsilon_E}{\hat{E}_{n-1}^u + \varepsilon_E}. \quad (44)$$

The second-order velocity indicator is

$$\mathbf{A}_n^u = \frac{2}{\Delta t_n + \Delta t_{n-1}} \left(\frac{\mathbf{u}_n - \mathbf{u}_{n-1}}{\Delta t_n} - \frac{\mathbf{u}_{n-1} - \mathbf{u}_{n-2}}{\Delta t_{n-1}} \right), \quad (45)$$

$$\hat{\kappa}_n^u = \frac{T_{\text{ref}}^2 \|\mathbf{A}_n^u\|_{M_u}}{\sqrt{|\Omega|} U_{\text{ref}}}. \quad (46)$$

The six statistics $\mathbf{s}_n = [\hat{E}_n^u, \hat{E}_n^p, \hat{D}_n^u, \hat{D}_n^p, \hat{g}_n, \hat{\kappa}_n^u]^\top$ are standardized as

$$\mathbf{d}_n = (\mathbf{s}_n - \boldsymbol{\mu}_s) \oslash (\boldsymbol{\sigma}_s + \varepsilon_s \mathbf{1}) \in \mathbb{R}^6. \quad (47)$$

Because these quantities are computed from physical fields, they do not depend on either candidate's POD coordinates.

Correction gate and objective. The gate input $[\tilde{\mu}; \mathbf{d}_n] \in \mathbb{R}^7$ is processed by two hidden layers of width 64 with SiLU activations and a scalar linear output. The output layer is zero-initialized, so training begins from the E2 pair probability. The gate predicts

$$\delta = c_\vartheta([\tilde{\mu}; \mathbf{d}_n]), \quad \alpha = \text{sigmoid}(\ell^{E2} + \delta). \quad (48)$$

Both specialists start from the same three-state physical history, encode it in their native charts, and perform independent autonomous rollouts. The scalar α is fixed over the prediction window. Velocity and gauge-corrected pressure are reconstructed on the common mesh and combined convexly; the fused output is never projected back into either rollout.

For candidate field errors e_1^c and e_2^c , $c \in \{u, p\}$, define

$$Q^c(\alpha) = \alpha^2 \|e_1^c\|_{M_c}^2 + (1 - \alpha)^2 \|e_2^c\|_{M_c}^2 + 2\alpha(1 - \alpha) \langle e_1^c, e_2^c \rangle_{M_c}. \quad (49)$$

The correction map minimizes

$$\mathcal{L}_{\text{T2C}} = \text{mean}_{w,k} \left[\frac{Q^u(\alpha^{(w)})}{\|\mathbf{u}_{\text{true}}\|_{M_u}^2 + \varepsilon} + \frac{Q^p(\alpha^{(w)})}{\|p_{\text{true}}^o\|_{M_p}^2 + \varepsilon} \right]. \quad (50)$$

Gate checkpoints are selected on validation data by lexicographically minimizing the worst Reynolds-number mean error and then the global mean error. The equal blend fixes $\alpha = 1/2$; the probability blend sets $\alpha = \bar{\pi}_1$; and the convex oracle chooses its weight a posteriori for each evaluation window. The oracle is a reference, not a deployable method.

E REPRODUCIBILITY AND IMPLEMENTATION DETAILS

Trajectory-level data partition and evaluation protocol. The split unit is a complete Reynolds-number trajectory, not an individual snapshot or a randomly sampled time window. Each local specialist is fitted only on its stated training trajectories, model and routing choices use the validation trajectories, and test trajectories are used only for the reported autonomous rollouts. A Reynolds number that occurs in two overlapping fluidic-pinball charts keeps the same split role in both.

Table 5: Chart-wise parameter coverage, trajectory counts, and rollout horizons. Train, validation, and test counts refer to complete Reynolds-number trajectories.

Benchmark	Chart	Re range	Train	Validation	Test	K
Circular cylinder	S	20.0–46.4	14	2	4	56
	H	45.5–59.2	29	2	3	56
	P	57.3–200	53	6	4	48
Centered square	S	50–95.4	60	5	4	16
	H	94–102	23	6	5	48
	P	98.5–150	27	6	4	24
Fluidic pinball	S	10–19	37	8	7	56
	H	16.6–25	39	8	7	56
	P	21.25–60	47	9	9	32

Unified autonomous integrator. All steady, Hopf, and periodic specialists use the same classical fourth-order Runge–Kutta update with a frozen context within each macro-step. Specifically, the pressure coefficients, recent-history features, and physical parameter are held fixed across the four RK stages; only the trial velocity state changes from stage to stage. After the RK4 velocity update, the algebraic pressure closure is evaluated, the history is shifted, and the resulting predictions are fed back at the next macro-step. The rollout is initialized from the available physical history, and no reference state from inside the prediction window is injected thereafter.

Datasets and local reduced dimensions. For the circular cylinder, all three charts use $(r_u, r_p) = (32, 32)$ and their regime-specific snapshot intervals. For the centered square, (r_u, r_p) is $(5, 4)$, $(11, 11)$, and $(28, 26)$ for the steady, Hopf, and periodic charts, respectively. Its 120 complete trajectories cover $Re \in [50, 150]$ on a 9,400-cell finite-volume mesh; the CFD step is $\delta t_{\text{CFD}} = 0.02$, and one snapshot is recorded every 200 CFD steps, so $\Delta t_{\text{snap}} = 4$. For fluidic pinball, the corresponding ranks are $(4, 3)$, $(5, 5)$, and $(17, 16)$. Its 131 complete trajectories cover $Re \in [10, 60]$, with 94/19/18 train/validation/test trajectories overall, on a 70,194-cell mesh; $\delta t_{\text{CFD}} = 0.005$ and snapshots are sampled every 0.25 physical time units. Chart-wise Reynolds-number ranges, split counts, and rollout horizons are listed in Table 5.

Admissible routing and fusion. The outer regime graph permits only adjacent S – H and H – P pairs; neither a direct S – P pair nor a three-specialist route is evaluated. The parameter-only E2 probabilities are evaluated once for a complete fixed- Re trajectory and remain fixed throughout its

rollout. In an admissible Top-2 case, the two specialists encode the same initial physical history, evolve independently in their native coordinates, and are combined only after reconstruction in the common physical space. Rollout windows and routing/fusion choices are fixed using validation data before test evaluation.

Reproducibility limitations. The experiments use fixed validation-selected training runs; repeated-seed dispersion and parameter-matched capacity controls are not evaluated.

F CIRCULAR-CYLINDER RESULTS

Table 6 reports the min–max range over held-out long-horizon trajectories. “N/E” denotes not evaluable under the prescribed rollout criterion.

Table 6: Circular-cylinder long-horizon error ranges (%). The Vanilla-FNN-MoE Hopf model does not meet the validation-stage stability requirement and is therefore not evaluated on held-out trajectories (N/E).

Regime	Model	E_u	E_p
Steady	RAL-MoE-ROM	0.0330–0.1723	0.2500–1.3249
	Vanilla-FNN-MoE	0.0384–0.2995	2.2163–18.5603
	DataOnly-MoE	0.1637–0.5122	4751.1025–27275.4645
	Global MoE	0.0845–0.3404	1.5226–7.4363
Hopf	RAL-MoE-ROM	0.0032–0.0288	0.0355–0.4321
	Vanilla-FNN-MoE	N/E	N/E
	DataOnly-MoE	0.1260–0.3880	1280.2811–1524.5949
	Global MoE	0.0110–0.0268	0.0560–0.2882
Periodic	RAL-MoE-ROM	0.3809–1.3473	1.6825–5.9601
	Vanilla-FNN-MoE	1.0393–3.0836	3.9659–13.3257
	DataOnly-MoE	0.7854–1.8677	2.2511–10.8189
	Global MoE	0.5873–1.1726	2.1896–6.3566

In the S – H overlap, E2 Top-1 gives $E_u = 0.0357$ – 0.0410% and $E_p = 1.6746$ – 1.9028% . T2-C reduces these ranges to 0.0148 – 0.0355% and 0.7387 – 1.7212% , with mean steady weight spanning 0.0861 – 0.9987 . The global model yields $E_u = 0.0851$ – 0.0953% and $E_p = 1.9517$ – 2.3106% on the same overlap evaluation.

G CENTERED-SQUARE RESULTS

Table 7 separates the velocity and pressure errors for the joint-field comparison in Table 3. Best single in the main text means one fixed specialist per overlap, chosen for comparison by the lowest aggregate joint error: Hopf for S – H and Periodic for H – P . It is distinct from the window-wise convex oracle, which uses the reference solution to choose a weight for each evaluation window.

The aggregate local-specialist errors are $(E_u, E_p) = (1.5311, 1.4583)\%$ for steady flow, $(0.4387, 0.4844)\%$ for Hopf-onset flow, and $(0.9395, 1.0921)\%$ for periodic flow. On the aggregate S – H evaluation, the steady and Hopf specialists have joint errors 4.8426% and 1.7298% ; on H – P , the periodic and Hopf specialists give 1.2079% and 2.8718% . These results show why the best local choice can change between overlaps and why a corrected convex assembly is useful.

Table 7: Complete centered-square routing and fusion errors at $K = 24$. All errors are percentages. The convex oracle is an a posteriori reference; bold marks the best deployable assembly for each field metric.

Overlap	Predictor	E_u	E_p	E_{joint}
$S-H$	Steady specialist	2.5976	2.2450	4.8426
	Hopf specialist	0.4942	1.2356	1.7298
	E2 Top-1	0.4942	1.2356	1.7298
	Equal blend	1.3644	1.6677	3.0321
	E2-probability blend	0.7898	1.3949	2.1847
	T2-C	0.4615	0.7343	1.1958
	Convex oracle	0.4659	0.7183	1.1842
$H-P$	Periodic specialist	0.6420	0.5659	1.2079
	Hopf specialist	1.0034	1.8684	2.8718
	E2 Top-1	0.7422	1.1052	1.8473
	Equal blend	0.6077	1.0069	1.6146
	E2-probability blend	0.6154	1.0464	1.6618
	T2-C	0.4509	0.3956	0.8465
	Convex oracle	0.4479	0.3905	0.8384

Table 8: Parameter-wise centered-square routing and T2-C results at $K = 24$. Joint errors and relative error reductions are percentages. The mean weight is the steady weight for $S-H$ and the periodic weight for $H-P$.

Overlap	Re	E2 choice	$\bar{\alpha}$	E2 E_{joint}	T2-C E_{joint}	Reduction
$S-H$	95.1	Hopf	0.0967	1.7140	1.1827	31.00%
$S-H$	95.3	Hopf	0.0971	1.7456	1.2090	30.74%
$H-P$	100.5	Hopf	0.5870	2.6485	0.8571	67.64%
$H-P$	102.0	Periodic	0.7094	1.0462	0.8360	20.09%

Figures 3 and 4 provide the complete velocity, gauge-corrected pressure, and pointwise-error views underlying the compact main-text comparison.

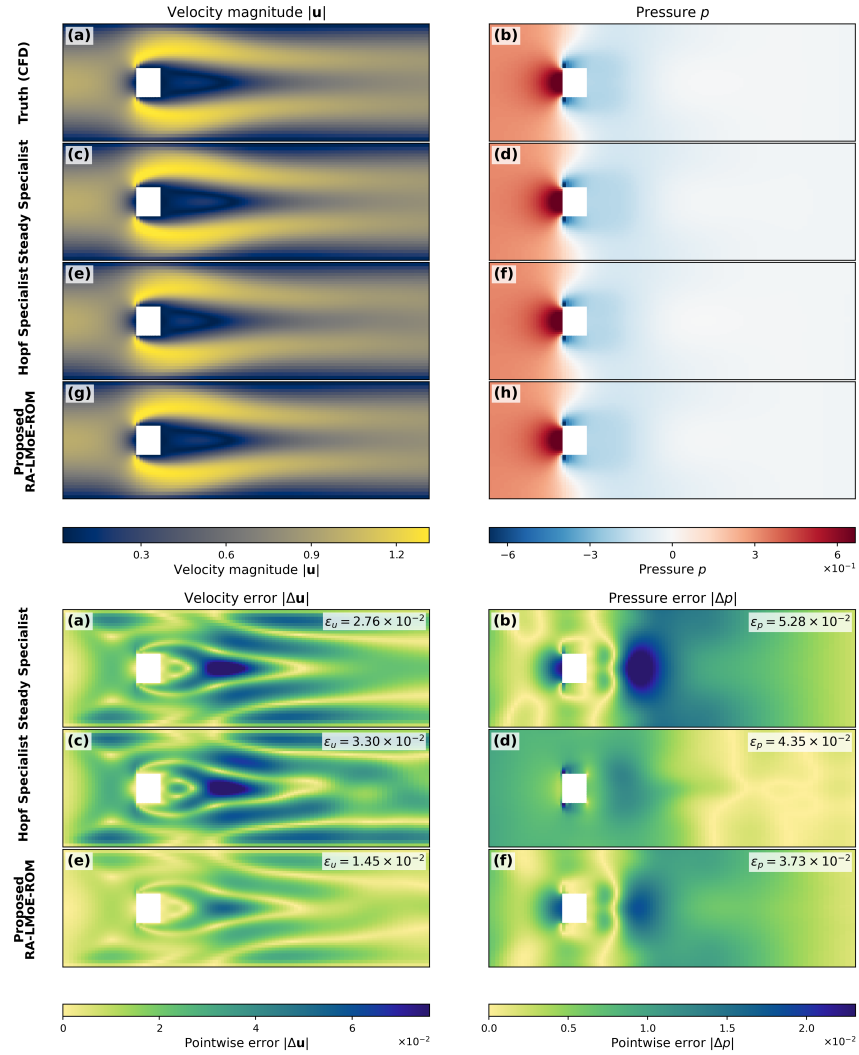


Figure 3: Representative held-out prediction in the centered-square $S-H$ overlap at $Re = 95.1$. The upper panel compares the CFD reference, Steady specialist, Hopf specialist, and T2-C prediction for velocity magnitude and gauge-corrected pressure. The lower panel gives the corresponding pointwise absolute errors.

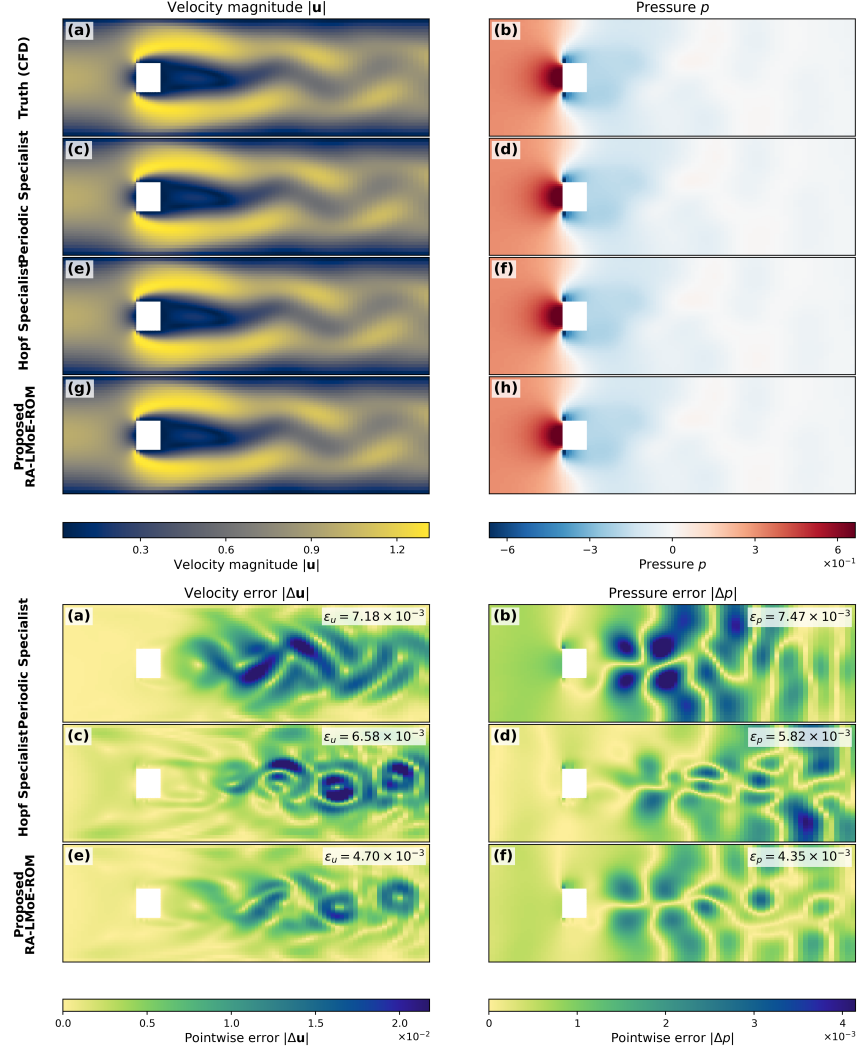


Figure 4: Representative held-out prediction in the centered-square H - P overlap at $Re = 100.5$. The upper panel compares the CFD reference, Periodic specialist, Hopf specialist, and T2-C prediction for velocity magnitude and gauge-corrected pressure. The lower panel gives the corresponding pointwise absolute errors.

H FLUIDIC-PINBALL RESULTS

The fluidic-pinball dataset contains 131 trajectories over $Re \in [10, 60]$, with 94/19/18 train/validation/test trajectories overall. A POD-DeepONet one-step baseline gives $E_u = 1.5921\%$ and $E_p = 23.3447\%$ on 4,228 held-out input-output pairs, but only 1 of 28 held-out trajectories remains finite under recursive rollout. This contrast motivates evaluation under autonomous use rather than one-step accuracy alone.

Table 9: Fluidic-pinball overlap joint errors (%). Primary denotes the Steady specialist for $S-H$ and the Periodic specialist for $H-P$; the adjacent specialist is Hopf in both cases. The oracle selects its convex weight a posteriori for each evaluation window and is unavailable during prediction.

Overlap	K	Primary	Adjacent	E2 prob.	T2-C	Oracle
$S-H$	24	0.6014	3.1925	1.7511	0.5897	0.5889
$S-H$	56	0.6927	3.1193	1.7714	0.6767	0.6763
$H-P$	24	0.0990	0.5146	0.2972	0.0945	0.0928
$H-P$	56	0.1028	0.5346	0.3085	0.0978	0.0955

T2-C improves the primary single-specialist prediction in all four overlap evaluations and remains close to the window-wise convex oracle. The result is consistent across the short and long rollout horizons shown here.